

Adaptive Multimodal Human-Robot Communication Using a Multi-Agent AI Framework with Safety-Aware Decision Making

Problem Identification - Domain and Motivation:

Application Domain

This topic falls under the domain of Human–Robot Interaction (HRI), specifically focusing on adaptive multimodal communication in collaborative robots (cobots). Human–robot collaboration is increasingly being adopted in healthcare, assistive living, manufacturing, education, logistics, and service robotics, where robots work alongside humans instead of operating in isolation. In these environments, robots must understand natural human communication through multiple modalities such as speech, gestures, facial expressions, and visual cues while making safe and context-aware decisions.

Advances in artificial intelligence and large language models (LLMs) have significantly improved robots' reasoning and language capabilities. However, most existing systems still rely on single-modal interaction or task-specific architectures, limiting their ability to communicate naturally and adapt to dynamic environments. Developing adaptive multimodal communication frameworks is therefore essential for enabling intuitive, efficient, and trustworthy human–robot collaboration.

Motivation

The importance of intelligent human-robot collaboration is driven by the growing demand for assistive technologies and collaborative robotic systems drives robot collaboration. According to the World Health Organisation (WHO), more than 2.5 billion people worldwide currently require one or more assistive products, and this number is projected to exceed 3.5 billion by 2050 due to population ageing and the increasing prevalence of chronic diseases. Assistive technologies, including intelligent robotic assistants, improve independence, education, employment opportunities, and overall quality of life for older adults, people with disabilities, and individuals with long-term health conditions.

Furthermore, the WHO highlights that assistive technologies benefit not only end users but also caregivers and healthcare systems by reducing the burden of long-term care, improving rehabilitation outcomes, and enabling more efficient healthcare delivery.

Beyond healthcare, collaborative robots are increasingly being adopted in manufacturing, logistics, warehouses, smart homes, retail, hospitality, agriculture, and public service environments. Unlike traditional industrial robots that operate in isolated workspaces, collaborative robots are designed to work alongside humans, requiring natural, intuitive, and safe interaction. As these robots become part of everyday environments, they must be capable of understanding diverse human communication methods and responding appropriately in dynamic situations.

Effective human–robot interaction (HRI) relies on accurate interpretation of multimodal human inputs, including speech, hand gestures, facial expressions, body posture, visual context, and environmental information. Relying on a single communication modality often leads to errors due to noisy environments, speech ambiguity, occluded objects, poor lighting, or unclear user intent. These limitations reduce collaboration efficiency and user confidence.

Another critical challenge is operational safety. Many existing robotic systems execute commands immediately after recognition without verifying whether the requested action is safe. This can result in accidents such as collisions with humans, picking hazardous objects, or operating in unsafe environments. Safety-aware reasoning is therefore essential before action execution, particularly in healthcare, rehabilitation, and collaborative industrial settings where robots operate in proximity to people.

Recent advances in Artificial Intelligence, computer vision, speech recognition, and multi-agent systems provide an opportunity to address these challenges. By combining information from multiple intelligent agents, robots can better understand user intent, resolve ambiguities, evaluate environmental risks, and make more reliable decisions. This motivates the development of a multimodal, multi-agent AI framework with safety-aware decision-making that improves interaction accuracy, enhances user trust, and enables safe deployment of collaborative robots in real-world environments.

Stakeholders

The proposed system is intended to support multiple stakeholders:

- **Primary stakeholders**
 - Older adults requiring assistance with daily living activities and independent living.
 - People with disabilities who rely on assistive robotic technologies for mobility, communication, or rehabilitation.
 - Patients receiving rehabilitation or long-term healthcare support.
 - Healthcare professionals (doctors, nurses, physiotherapists, occupational therapists) using collaborative robots during patient care.
 - Industrial workers collaborating with robots in manufacturing, assembly lines, warehouses, and logistics.
 - Service personnel in hotels, restaurants, airports, and retail environments using robots for customer assistance.
 - Robot operators and technicians responsible for supervising, monitoring, and maintaining robotic systems.
 - Home users interacting with domestic assistant robots for household support.
- **Secondary stakeholders**
 - Robotics researchers developing improved Human–Robot Interaction algorithms.
 - AI and machine learning developers designing multimodal perception and decision-making systems.
 - Healthcare institutions, hospitals, rehabilitation centres, and elderly care facilities deploying assistive robots.

- Manufacturing companies implementing collaborative robots to improve productivity and worker safety.
- Logistics and warehouse operators adopting autonomous collaborative robots.
- Caregivers and family members who supervise or assist robot users.
- Educational institutions and research laboratories using collaborative robots for teaching and experimentation.
- Government agencies and regulatory bodies responsible for robotic safety standards, AI governance, and certification.
- Robot manufacturers and system integrators designing and commercialising collaborative robotic platforms.
- Insurance providers and workplace safety organisations concerned with reducing occupational accidents and ensuring compliance with safety regulations.

Decision Supported by the System

The proposed system supports real-time, multimodal, and safety-aware human–robot decision-making through collaboration among multiple specialised AI agents. The robot is capable of:

- Processing multimodal human inputs by recognising speech, hand gestures, and visual information through dedicated Voice, Gesture, and Vision Agents.
- Combining information from multiple modalities using the Coordinator Agent to accurately infer the user's intended command and resolve ambiguities between different inputs.
- Utilising contextual memory through the Memory Agent to retain relevant interaction history, user preferences, and previous tasks, enabling more personalised and consistent robot behaviour.
- Planning appropriate robot actions using the Task Planner, which generates an execution strategy based on the inferred user intent and the current environmental context.
- Evaluating safety before execution through the Safety Agent by detecting obstacles, unsafe object manipulation, or hazardous situations and approving only safe actions.
- Executing validated robot actions through the Robot Controller after successful task planning and safety verification.

Unlike conventional human–robot interaction systems that rely on predefined commands or a single communication modality, the proposed multi-agent framework integrates multimodal perception, contextual memory, task planning, and safety-aware reasoning to support adaptive, reliable, and human-centred decision-making in dynamic real-world environments.

Literature Survey

Ref	Year	Dataset	Method/Architecture	Key Metric and value	Limitations
[1]	2026	5-batten physical timber structures & 26-batten RViz simulation	Multi-agent GPT-4o + RAG + MuJoCo + ROS/MoveIt for Human–Robot Collaboration	Demo I: 70 s; Demo II: 220 s; Demo III: Partial success (26-element simulation only); physics-based decision validation	Hallucination on complex task, no standard dataset, limited to lab environment, communication ambiguity, slow execution.
[2]	2026	5-object simulation (36 trials/object) + 20 real-world trials on LeeAhn 2	Whisper-small + GPT-4o Voice-to-Action with constrained skill grounding and safety validation	Seeking: 100% (simulation), 80% (real); Navigation: 72.2–97.2% (simulation), 65% (real); Manipulation: 78% (simulation avg.), 55% (real)	Single robot platform, latency, predefined skill library, AR-tag dependency, limited real-world evaluation.
[3]	2026	NetLogo simulation (2–10 agents) validated using Hancock et al. (2021) HRI meta-analysis	Agent-Based Model (ABM) for trust-aware human–robot team performance prediction	50% interval validity; strong ordinal validity; robot reliability was the strongest predictor of trust and performance; significant effects of reliability, transparency, communication, and collaboration	Simulation only; simplified trust model; no real robot experiments; depends on meta-analysis-based validation
[4]	2026	Real-time video captured using Raspberry Pi camera (indoor and outdoor object detection scenarios)	Multi-agent framework consisting of Vision (YOLOv8), Reporting (Ollama LLM), Communication (Slack), Control Agent, and Core Router on Raspberry Pi	Successfully achieved local multi-agent object detection and natural language reporting; YOLOv8n delivered the best detection/tracking performance; full local system operation confirmed	Raspberry Pi resources were insufficient for simultaneous YOLO and LLM inference, causing high CPU load, <3 FPS, notification delays, and occasional report timeouts
[5]	2025	Two conceptual use cases: Satellite & Swarm Robotics and Large Action Models (LAMs) for Human-AI collaboration	System-theoretical Human-AI framework combining Multi-Agent Systems (MAS), Centaurian systems, Communication Spaces, and Colored Petri Nets	Introduced a unified architecture for adaptive Human-AI interaction with layered communication (Surface–Observation–Computation), human-in-the-loop decision making, feedback loops, and formal coordination suitable for robotics and intelligent	Primarily theoretical with illustrative case studies; lacks experimental validation, quantitative performance metrics, and real-world safety evaluation

				systems	
[6]	2013	Vision-based hand gesture recognition for HCI, robotics, VR, sign language, and consumer electronics	Classical computer vision pipeline: hand detection, segmentation, tracking, feature extraction, and classification using HMM, ANN, SVM, and template matching	Comprehensive survey of traditional vision-based HGR methods, taxonomy of gesture recognition systems (data glove, colour glove, vision-based), and analysis of recognition pipeline and applications	Does not include deep learning, Vision-Language Models, LLMs, multimodal reasoning, semantic understanding, or adaptive human–robot communication; performance is sensitive to lighting, occlusion, and background complexity.
[7]	2025	Indoor mobile manipulator performing spoken search and manipulation tasks	LLM-based Voice-to-Action framework combining speech recognition, structured LLM task planning, predefined robot skill library, navigation, vision, and manipulation	Demonstrates practical voice-controlled robot execution with constrained LLM-generated action plans, failure recovery, and validation in simulation and real-world experiments	Voice-only interaction; no multimodal fusion, multi-agent collaboration, adaptive personalisation, or comprehensive safety-aware reasoning; LLM latency and occasional execution failures remain.
[8]	2024	Simulated robotic environments (class, pose, partial observability, physical, and SLAM uncertainty) and real-world Franka robot experiments for object search and safe human–robot interaction	TAMPURA (Task and Motion Planning with Uncertainty and Risk Awareness): belief-state abstraction, Bayesian model learning, sparse MDP construction, LAO*-based probabilistic planning	Enables uncertainty-aware and risk-aware long-horizon planning, learns transition probabilities online, supports information gathering, and outperforms RL, MCTS, and deterministic TAMP baselines in uncertain environments	Requires manually designed belief abstractions and symbolic operators, computationally intensive, lacks multimodal communication, LLM integration, adaptive personalisation, and multi-agent collaboration.
[9]	2025	COOLER (extended COOL benchmark): 200 traffic videos with denoising, deblurring, human hazard annotations, and open-set	Multi-agent Vision-Language framework using OmniVLM, ViLA, GPT-4o-mini, CLIP, and two-track hazard reasoning for zero-shot hazard detection	BESM = 0.3922, SAM = 0.3922; semantic evaluation using cosine similarity (>0.80 threshold) for hazard descriptions; detects and localizes novel hazards through zero-shot reasoning	Difficulty detecting small or occluded hazards, failures in multi-agent track merging, computationally intensive, and evaluated only on hazard captioning rather than complete autonomous driving tasks.

		evaluation			
[10]	2025	Kaiwu Kitchen Dataset: 20 participants, 17 kitchen tools, 30 task scenarios, 680 task segments (~11 hours), 7 sensing modalities, 56,000 annotations	Multimodal human demonstration dataset integrating RGB-D, motion capture, EMG, eye tracking, tactile gloves, audio, and full-body sensing for robotic skill acquisition	680 task segments, ~11 hours of synchronised recordings, 7 modalities, 56,000 annotations, 20 participants, 17 manipulation tasks; provides comprehensive multimodal benchmark for robotic manipulation and embodied AI	Limited to kitchen tasks, small participant pool, focuses on dataset creation rather than adaptive HRI, lacks LLM integration, multi-agent AI, and safety-aware decision-making.
[11]	2026	Timber assembly, 7 participants, Human–Robot Collaboration (HRC)	Multimodal Agentic AI, Whisper, Llama 3.2, Gaze Tracking, ROS, Apriltag	18.6% intervention rate (vs. 37.1% baseline), Lower NASA-TLX workload, Improved HRC efficiency	Small sample size, Lab-based validation, Limited robot actions, High latency, No real-world object detection.
[12]	2024	Overcooked-AI, 16 participants, Easy/Medium/Hard task layouts	HRT-ML, GPT-4o, Coordinator–Manager architecture, Adaptive multimodal language feedback	Higher collaboration performance, Improved trust & perceived intelligence, Active feedback effective for complex tasks, Passive feedback better for simple tasks	Simulation-based evaluation, Small participant group, Static feedback levels, No real-world robot validation.
[13]	2026	2,400 indoor images, LLM-SAP, Hazards&Robots, Home Fire Dataset	MARS, Multi-Agent MLLM, CLIP + DeepLabV3, Risk Assessment, Planning & Evaluation Agents	Superior risk-aware planning, improved multi-agent coordination, Best overall ranking compared to GPT-4o, Qwen2.5-VL, and LLaVA	Static image-based evaluation, Indoor assistive scenarios only, no real-world robot deployment, requires high computational resources.
[14]	2025	Industrial assembly simulation, MATLAB R2023b, Python 3.10, 10 agents (5 humans, 5 robots), 10 task levels	Game-Theoretic Decision Engine, Multimodal Perception (Vision + Audio + Tactile), Sensor Fusion, Human Intention Recognition, System Configuration Optimiser	Efficiency improved from 0.85 → 0.98 (~15%), Collaborative system maintained 5–8% higher efficiency than standalone human/robot systems, 100% task completion within 60 minutes	Dependent on high-quality sensor data, Performance affected by lighting, noise, and environmental variability, assumes rational human decision-making, Scalability to larger industrial environments remains challenging, Limited long-term adaptability without reinforcement learning.
[15]	2022	Escape Room (Mixed Reality) Scenario, Pepper Robot,	Co-adaptive Human–Robot Framework, Meta-Learning +	Perceived trust significantly improved with meta-learning: $F(1,22)=16.44, p<0.001$,	Requires high-quality multimodal sensing and user feedback, Computational complexity

		Microsoft HoloLens, 30 participants (presentation study), Collaborative packaging task using body-motion data	Reinforcement Learning (MAML + TRPO), Multi-Armed Bandit (Exp3), Behaviour Trees, Knowledge Graphs, Temporal Logic (LTL & MITL), Mixed-Initiative Control	$\eta^2=0.428$; Adaptive presentation system preferred by users over the non-adaptive version; Safety guaranteed through mixed-initiative control while adapting to human preferences.	due to temporal logic and reinforcement learning, Limited validation in controlled laboratory scenarios, Scalability to real-world industrial environments not fully evaluated, Long-term adaptation and multi-user collaboration remain open challenges.
[16]	2025	ELDERLY-AT-HOME corpus, 12 healthy adults, 75 real-world trials, multimodal human-robot collaboration dataset (speech, pointing gestures, haptic-ostensive actions)	Deep Q-Learning (DQN), DAGGER warm-up, Reinforcement Learning, Neural User Simulator, ALBERT-based Dialogue Act Classifier, Multimodal Interaction Manager, Baxter Robot	Overall system accuracy: 90.2%; RL interaction manager accuracy: 97.5%; Real-time RL policy accuracy: 98.1%; Dialogue Act classifier accuracy: 88.41%; User simulator overall accuracy: 70.85% (State accuracy 89.44%, Action accuracy 77.89%, DA accuracy 75.38%). The RL policy outperformed the previous HBATN rule-based interaction manager while maintaining high task success and user satisfaction.	Evaluated only on a single "Find" task, small user study (12 participants), limited object/location vocabulary, requires accurate speech and gesture recognition, performance depends on simulator quality, tested only in a controlled indoor environment, and generalisation to more complex collaborative tasks remains future work.
[17]	2026	105 participants, 2 NAO robots, 7 experimental blocks	M2HRI, GPT-4o-mini, VLM, Big Five Personality, Long-term Memory, LLM-based Coordinator	93.3% personality recognition, improved recall, personalisation, conversational flow, and overlap avoidance	Video-based study, 2 robots only, no long-term real-world evaluation, no adaptive personality learning.
[18]	2024	Overcooked-AI, 16 participants, 192 trials, Easy/Medium/Hard layouts	HRT-ML, GPT-4o, Coordinator-Manager Framework, Adaptive Multimodal Language Feedback	Higher trust & perceived intelligence, Improved team performance, Active feedback effective for complex tasks, Passive feedback better for simple tasks	Simulation-based study, Small sample size, Static feedback levels, No real-world robot validation.
[19]	2026	2 NAO humanoid robots, Human-Multi-Agent interaction, Demonstration scenarios	Multimodal HRI Framework, VLM + LLM, Centralised Coordinator, Speech, Gesture, Gaze & Locomotion	Coordinated turn-taking, Multimodal reasoning, Grounded embodied responses, Reduced speech overlap & action conflicts	Demonstration-based evaluation, no large-scale user study, Limited to two robots, Scalability and latency challenges.
[20]	2025	Literature review of EMAS,	Systematic Review, Generative	Comprehensive taxonomy, Unified EMAS framework,	Survey paper (no experimental validation), no

		AI2-THOR, Habitat, Minecraft, Overcooked-AI, BEHAVIOR-1K, Grid World	Multi-Agent Collaboration, Foundation Models (LLMs/VLMs), EMAS Taxonomy, Perception–Planning–Communication–Feedback Framework	Highlights FM-based collaboration, Identifies key applications & future research directions	quantitative performance metrics, Limited by existing literature, Standardised benchmarks still lacking.
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Problem Statement

Existing human–robot interaction systems primarily rely on single-modal communication or task-specific architectures, limiting their ability to interpret diverse human inputs and adapt to dynamic collaborative environments. The proposed system consumes multimodal inputs, including speech, hand gestures, facial expressions, visual information, and environmental context, to accurately infer user intent and generate a safety-aware execution plan through a multi-agent AI framework. The primary challenge lies in handling noisy and ambiguous multimodal data, resolving conflicting information across modalities, maintaining real-time response, and ensuring safe operation during close human–robot collaboration. The success of the proposed system will be evaluated by comparing its human intent recognition accuracy, task completion rate, response latency, safety performance, and user interaction quality against existing single-modal and conventional human–robot communication approaches.

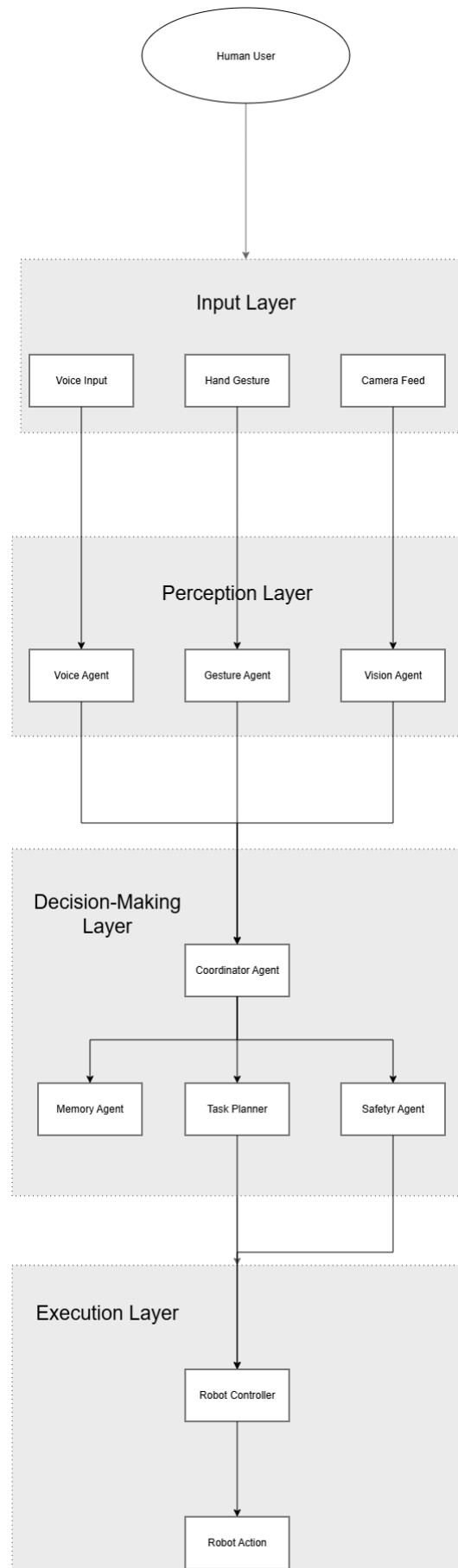
Gap analysis:

The literature review indicates significant progress in individual aspects of Human–Robot Interaction (HRI), including voice-based interaction, gesture recognition, multimodal perception, large language model (LLM)-based task planning, safety-aware motion planning, and multi-agent AI systems. However, most existing studies focus on only one or two of these capabilities.

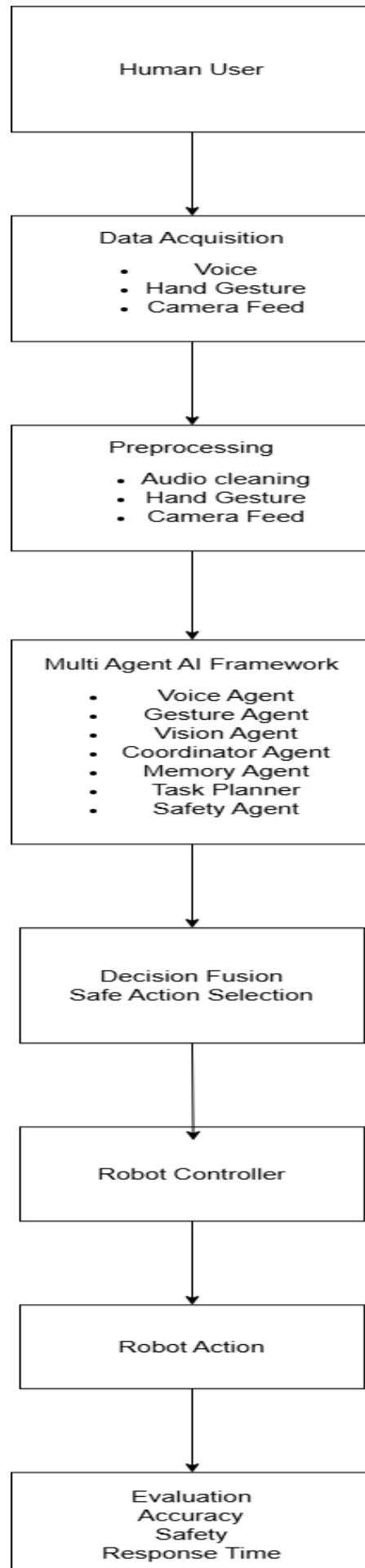
Many voice-based systems rely solely on speech commands and do not integrate complementary communication modalities such as hand gestures or visual context. Similarly, gesture recognition systems often perform only perception tasks without supporting high-level reasoning or adaptive decision-making. Although recent multi-agent frameworks improve task coordination, they generally lack multimodal human communication and personalised contextual memory. Existing safety-aware planning methods primarily address navigation and motion planning rather than validating multimodal user intentions before robot execution.

Furthermore, very few studies combine multimodal perception, contextual memory, task planning, safety verification, and multi-agent reasoning into a unified architecture for real-time human–robot collaboration. Most systems are also evaluated in controlled laboratory environments with limited adaptability to dynamic real-world scenarios. Therefore, there remains a need for an adaptive multimodal human–robot communication framework that integrates multiple AI agents, contextual memory, multimodal intent understanding, and safety-aware decision-making to enable reliable, efficient, and human-centred collaboration in complex environments.

Proposed System Architecture



Pipeline Diagram:



Justification for Every Design Choice

Component	Justification
Voice Agent	Speech is the most intuitive communication modality in HRI but becomes unreliable in noisy environments; therefore, it is combined with other modalities.
Gesture Agent	Gestures provide complementary information when speech is ambiguous or unavailable.
Vision Agent	Visual context enables identification of objects and environmental conditions required for task execution.
Coordinator Agent	Multimodal fusion reduces ambiguity by integrating complementary information from multiple perception agents.
Memory Agent	Stores previous interactions to improve contextual understanding and consistency across tasks.
Safety Agent	Prevents unsafe robot actions by verifying environmental conditions before execution.
Task Planner	Converts user intent into an executable sequence of robot actions.

Planned Experimental Setup

The proposed framework will be evaluated using publicly available datasets for speech, hand gesture, and object recognition, supplemented by a small custom dataset to simulate multimodal human–robot interactions. The data will be divided into 70% training, 15% validation, and 15% testing. Performance will be assessed using accuracy, precision, recall, F1-score, task success rate, response time, and safety violation rate. The proposed framework will be compared against voice-only, gesture-only, and vision-only baseline systems, as well as a multimodal system without safety-aware decision-making. Experiments will be conducted on a workstation equipped with a modern multi-core CPU, GPU, and at least 16 GB RAM. Ablation studies will be performed by removing individual components, such as the Memory Agent, Safety Agent, or Coordinator Agent, to evaluate the contribution of each module to the overall system performance.

Feasibility Note

Compute Resources

The proposed system will be developed using Python, PyTorch, ROS2, and OpenCV. Model training will utilise a workstation equipped with an Intel Core i7 processor, 16 GB RAM, and an NVIDIA RTX 4060 GPU. Lightweight components such as Whisper-small, MediaPipe, and YOLO are selected to ensure feasible training and inference within the available hardware resources.

Dataset Availability

Publicly available datasets, including Mozilla Common Voice (speech), HaGRID or EgoGesture (gesture recognition), and COCO (object detection), will be used. Additional multimodal interaction samples may be collected for testing.

Estimated Development Time

- Data preparation: 2 weeks
- Model integration: 3 weeks
- Multi-agent framework development: 3 weeks
- Testing and evaluation: 2 weeks

Risks and Mitigation

Risk	Mitigation
Limited multimodal datasets	Combine multiple public datasets and collect a small custom dataset.
High inference latency	Use lightweight models and optimise inference.
Sensor failures or noisy inputs	Employ multimodal fusion so remaining modalities compensate.
Hardware limitations	Use smaller models and cloud resources if necessary.

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